

Section 5 — Initial Network

This section reports the **initial** facial recognition CNN (before hyperparameter optimization).

Architecture: Rescaling → Conv(16)-Pool → Conv(32)-Pool → Conv(64)-Pool → Conv(64)-Pool → Flatten → Dense(16) → Dropout(0.3) → Softmax(3). Optimizer: RMSprop (lr=0.001). No augmentation.

Total parameters: **110,755** (limit 150,000).

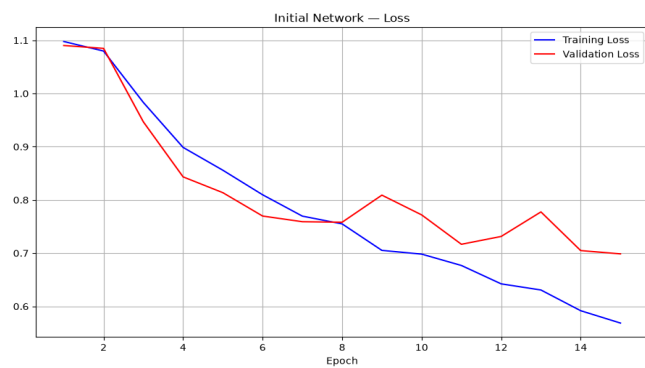
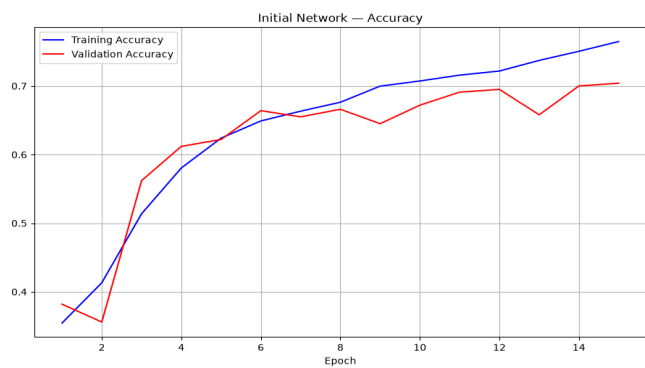
Epochs trained: **15**. Best validation accuracy: **0.7040**.

Held-out test accuracy: **0.7262**, test loss: **0.6855** (target ≥ 0.60). Saved model: **results/section5_initial_model.keras**.

Initial Network Summary

Model: "sequential"		
Layer (type)	Output Shape	Param #
rescaling (Rescaling)	(None, 150, 150, 3)	0
conv2d (Conv2D)	(None, 148, 148, 16)	448
max_pooling2d (MaxPooling2D)	(None, 74, 74, 16)	0
conv2d_1 (Conv2D)	(None, 72, 72, 32)	4,640
max_pooling2d_1 (MaxPooling2D)	(None, 36, 36, 32)	0
conv2d_2 (Conv2D)	(None, 34, 34, 64)	18,496
max_pooling2d_2 (MaxPooling2D)	(None, 17, 17, 64)	0
conv2d_3 (Conv2D)	(None, 15, 15, 64)	36,928
max_pooling2d_3 (MaxPooling2D)	(None, 7, 7, 64)	0
flatten (Flatten)	(None, 3136)	0
dense (Dense)	(None, 16)	50,192
dropout (Dropout)	(None, 16)	0
dense_1 (Dense)	(None, 3)	51
Total params: 110,755 (432.64 KB)		
Trainable params: 110,755 (432.64 KB)		
Non-trainable params: 0 (0.00 B)		

Initial Network Training Curves



Section 6 — Hyperparameter-Optimized Network

This section reports the **final optimized** facial recognition model used for the game controller and for all final accuracy numbers in this submission.

Architecture: Rescaling → RandomFlip → RandomRotation → RandomContrast → Conv(24)-Pool → Conv(32)-Pool → Conv(48)-Pool → Conv(64)-Pool → Dropout(0.25) → Flatten → Dense(32) → Dropout(0.4) → Softmax(3). Optimizer: Adam (lr=0.0009) with EarlyStopping + ReduceLROnPlateau.

Total parameters: **149,683** (limit 150,000).

Epochs trained: **35**. Best validation accuracy: **0.7860**.

Held-out test accuracy: **0.7775**, test loss: **0.5581** (target ≥ 0.70). Final saved model: **results/section6_final_model.keras**.

Optimized Network Summary

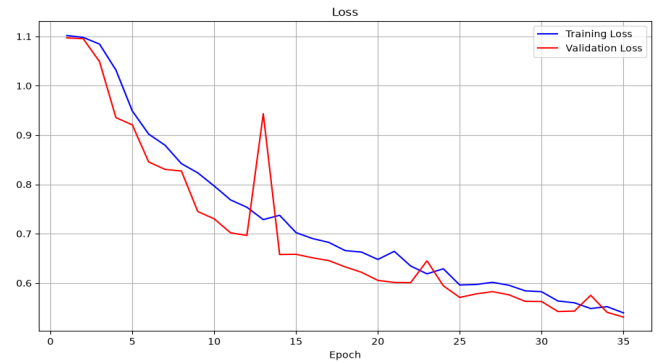
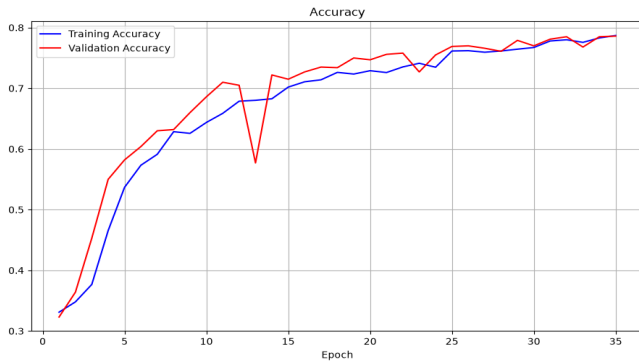
Model: "sequential"

Layer (type)	Output Shape	Param #
rescaling (Rescaling)	(None, 150, 150, 3)	0
random_flip (RandomFlip)	(None, 150, 150, 3)	0
random_rotation (RandomRotation)	(None, 150, 150, 3)	0
random_contrast (RandomContrast)	(None, 150, 150, 3)	0
conv2d (Conv2D)	(None, 148, 148, 24)	672
max_pooling2d (MaxPooling2D)	(None, 74, 74, 24)	0
conv2d_1 (Conv2D)	(None, 72, 72, 32)	6,944
max_pooling2d_1 (MaxPooling2D)	(None, 36, 36, 32)	0
conv2d_2 (Conv2D)	(None, 34, 34, 48)	13,872
max_pooling2d_2 (MaxPooling2D)	(None, 17, 17, 48)	0
conv2d_3 (Conv2D)	(None, 15, 15, 64)	27,712
max_pooling2d_3 (MaxPooling2D)	(None, 7, 7, 64)	0
dropout (Dropout)	(None, 7, 7, 64)	0
flatten (Flatten)	(None, 3136)	0
dense (Dense)	(None, 32)	100,384
dropout_1 (Dropout)	(None, 32)	0
dense_1 (Dense)	(None, 3)	99

Total params: 149,683 (584.70 KB)

Trainable params: 149,683 (584.70 KB)
Non-trainable params: 0 (0.00 B)

Optimized Network Training Curves



Hyperparameter Optimization Strategy

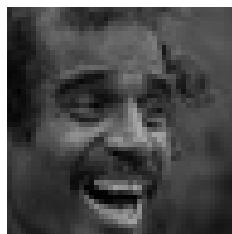
- Started from the Section 5 initial network (Dense 16, no augmentation, RMSprop).
- Increased Dense width 16 $\rightarrow$ 32 and conv widths to 24/32/48/64 while staying under 150k params.
- Added RandomFlip, RandomRotation, and RandomContrast for generalization.
- Inserted dropout after the last pool (0.25) and after Dense (0.4).
- Switched optimizer to Adam (lr=0.0009) and used EarlyStopping on val_accuracy.
- Result: test accuracy improved from 0.7262 (initial) to 0.7775 (optimized).

Example Training Images

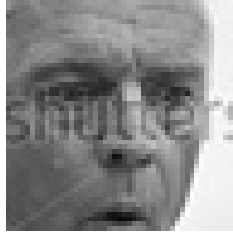
neutral



happy



surprise



Final model for submission / game: results/section6_final_model.keras with test accuracy 0.7775. Verify anytime with `python3 evaluate_model.py` (do not retrain).